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CivicWorkOS: Capability-Preserving Adaptive Task Allocation Among Humans, AI Agents, and Robots in Municipal Services

Toqeer Ali Syed[1,†], **Ali Akarma**[1,2,†], **Raghda M. Alqurashi**[3,*], **Muhammad Tayyab Naqash**[4], **Abdulaziz Alqurashi**[4]

[1]*AI Center, Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah, Saudi Arabia*
[2]*AI V&V Lab, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia*
[3]*Department of Computer Science, Umm Al-Qura University, Jumum, Saudi Arabia*
[4]*Civil Engineering Department, Islamic University of Madinah, Madinah, Saudi Arabia*

Correspondence*:
Raghda M. Alqurashi
rmqurashi@uqu.edu.sa

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ABSTRACT

Municipal services increasingly rely on humans, artificial intelligence (AI) agents, and robots to perform interdependent tasks. However, allocating work primarily to maximize immediate efficiency can reduce opportunities for human practice, weakening the skills needed for supervision, exception handling, and recovery from automation failures. This paper proposes CivicWorkOS, a capability-preserving framework for adaptive task allocation in municipal services. The framework treats human proficiency as a changing operational resource and jointly considers service quality, cost, safety, workload, and long-term human capability. An adaptive learning mechanism updates estimates of task requirements and participant capabilities using task outcomes, operational feedback, and multimodal observations where available. These estimates inform allocation decisions as service conditions, human proficiency, and robot performance change. Capability preservation is incorporated through supervised practice, collaborative execution, and periodic human-led tasks, while human oversight supports uncertain or consequential decisions. The framework also provides mechanisms for transferring control when an AI agent or robot becomes unreliable, with assignments constrained by human readiness and task-specific safety requirements. A reproducible evaluation protocol is defined to compare capability-preserving allocation against static and efficiency-focused approaches across routine operations, demand surges, and automation disruptions. Evaluation measures include task completion, service cost, workload, skill retention, and takeover performance. CivicWorkOS positions adaptive allocation as a means of improving municipal automation while sustaining the human expertise required for resilient service delivery.

Keywords: human–robot–AI task allocation, adaptive automation, capability preservation, skill retention, multimodal learning, control transfer, human–robot interaction, reproducible evaluation

[†]These authors contributed equally to this work and share first authorship.

1 INTRODUCTION

Municipal services are becoming mixed teams of people and machines. Computer-vision agents screen bridge imagery, language-model agents draft responses to citizen requests, and robots sweep streets, inspect pipes, and survey structures, while public employees supervise, decide, and repair what the machines cannot. Urban digital twins already run artificial intelligence (AI) agents over live municipal data (Choi and Yoon, 2025), and the resulting "agentic city" senses, learns, predicts, and acts (Shariatpour and Behzadfar, 2026). Each day such a city decides, task by task, which work a human performs, which an AI agent or robot performs, and which they perform together.

Those decisions are usually made for immediate efficiency, and efficiency has a blind spot. Human proficiency is not a fixed attribute. It grows through practice and decays without it (Arthur et al., 1998; Tatel and Ackerman, 2025), and the practice that sustains it is precisely the routine work automation absorbs first. Bainbridge (1983) named the consequence: automating the easy parts of a task leaves the human operator least practiced at the moment intervention is needed. A municipality that sends every routine inspection to a robot and every routine case to an AI agent will, a few years later, employ certified staff who cannot supervise the robot, catch the agent's error, or take over when either fails. Because each allocation is locally defensible, a task-by-task efficiency objective never registers the loss.

The problem is therefore one of adaptive allocation under a changing human state, and it has three parts that existing allocation methods treat separately, if at all. First, the allocator must *learn*: task requirements, practitioner proficiency, and machine reliability are latent, drift over time, and are observed only through outcomes and, increasingly, multimodal signals such as robot telemetry, skill probes, and physiological workload estimates (Dehais et al., 2020; Debie et al., 2021). Second, it must *preserve* capability, deliberately routing practice to developing practitioners, keeping certified ones current, and doing so within workload limits. Third, it must *transfer control* safely when an AI agent or robot becomes unreliable, which requires knowing which humans are ready to take over (Gold et al., 2013; Zhang et al., 2019). The data-driven treatment of these three parts, and their joint evaluation under a reproducible protocol, is the subject of this paper.

We present CivicWorkOS, a capability-preserving framework for adaptive task allocation among humans, AI agents, and robots in municipal services. Its decision variable is a *staffed mode*, an execution mode paired with a named roster of practitioners at declared career stages, so that allocation decides not only what kind of agent performs a task but which people practice on it. Its state includes a learned, uncertainty-aware estimate of every participant's current capability, and its constraints protect the practice through which human capability is formed and retained.

1.1 Contributions

- *Human proficiency as a changing resource.* A proficiency model in which credited practice raises a practitioner's attainable proficiency and gaps in human-led work erode current proficiency, with a closed-form recency interval (Proposition 1) that converts skill decay into a schedule of periodic human-led tasks.
- *An adaptive learning mechanism.* Gaussian beliefs over task requirements, practitioner proficiency, machine reliability, and workload response, updated by assumed-density filtering from task outcomes and by Kalman updates from multimodal observations where available. Every estimate carries an interpretable mean and variance, and the variance drives oversight.
- *Capability-preserving allocation.* A city-wide program and an online Lagrangian rule whose constraints implement supervised practice (a capability budget with a feasibility condition and a trainee-cohort requirement, Propositions 2 and 3), collaborative execution, periodic human-led tasks, a workload ceiling, and uncertainty- and consequence-gated human oversight.
- *Readiness-constrained control transfer.* A takeover rule that assigns control of a failing AI agent or robot only to a practitioner who is estimated to be current, has workload headroom, and can take over within the task's safety time budget, and otherwise commands a minimal-risk condition.
- *A reproducible evaluation protocol.* Three service domains, five policies (static, efficiency-focused, capability-preserving, and two ablations), three scenarios, five metric families, eight hypotheses fixed before execution, paired seeds, and an open pipeline, together with a reference execution over 30 seeds whose every reported number regenerates from one command.

- *An exact worked allocation.* A bridge-inspection instance computed end to end by arithmetic, which shows that machine assistance and shared supervision stretch time to competence from 1.2 to 4.05 years, so that a trainee cohort sized from certification hours provisions four posts where fourteen are required.

Section 2 positions the work. Sections 3 and 4 specify the framework and its allocation rule, Section 5 the evaluation protocol, Section 6 the worked allocation and the reference execution, and Section 7 what they do and do not establish.

2 RELATED WORK

Appendix S7 of the Supplementary Material extends this section work by work.

2.1 Task Allocation in Human–Robot and Human–AI Teams

Capability-based allocation assigns tasks by the fit between task requirements and agent capabilities (Ranz et al., 2017; Dorri et al., 2018). Later work added ergonomic risk (El Makrini et al., 2019; Merlo et al., 2023), human preferences (Noormohammadi-Asl et al., 2025), decision authority and worker satisfaction (Gombolay et al., 2015), and trust, either modeled explicitly (Ali et al., 2022) or inferred and planned with (Chen et al., 2018). Reviews cover human–robot collaboration (Ajoudani et al., 2018), robot fleets (Meseguer Valenzuela and Blanes Noguera, 2025), and multi-agent reinforcement learning for resource allocation (Hady et al., 2025), and the closest city-scale treatment assigns sensing tasks across a mobile crowd (Zhao et al., 2023). Personnel rostering carries skill mix, supervision ratios, and qualification maintenance as hard constraints (Burke et al., 2004; Van den Bergh et al., 2013), but assumes that the human-versus-machine decision has already been made. All of this work optimizes a team's present performance; none asks whether its allocations erode the human proficiency that future supervision and recovery depend on.

2.2 Adaptive Automation and Multimodal Human-State Estimation

Adaptive automation varies the level of automation with the operator's state and the task's demands (Parasuraman et al., 2000; Kaber and Endsley, 2004). Kaber and Endsley (2004) found that intermediate levels of automation preserve situation awareness that full automation loses, the out-of-the-loop problem of Endsley and Kiris (1995). Neuroergonomics supplies the measurement side (Parasuraman, 2003): electroencephalography, functional near-infrared spectroscopy, heart-rate variability, and eye tracking index mental workload and engagement (Dehais et al., 2020), multimodal fusion of these signals outperforms any single modality (Debie et al., 2021), and a passive brain–computer interface has triggered adaptive automation in realistic air-traffic control (Aricò et al., 2016). Subjective instruments such as NASA-TLX (Hart and Staveland, 1988) remain the reference. Trust calibration completes the picture (Lee and See, 2004; Wischnewski et al., 2023). These methods adapt automation to the operator's *momentary* state; CivicWorkOS adds the operator's *long-run* state, the proficiency that current allocation decisions build or erode.

2.3 Skill Acquisition, Retention, and Learner Modeling

Skill decays with the length of the non-practice interval, faster for cognitive than for physical skills (Arthur et al., 1998; Tatel and Ackerman, 2025), and learn–forget curve models capture this alternation of practice and interruption in production settings (Jaber and Bonney, 1996). Learner modeling estimates latent skill from noisy outcomes: Bayesian knowledge tracing (Corbett and Anderson, 1994) and its logistic successors (Pelánek, 2017) are the educational counterparts of the belief model of Section 3.4, and dynamic Bayesian rating systems (Glickman, 1999) update such beliefs online. Professional recency rules translate decay into schedules: a pilot must have flown recently to carry passengers (Federal Aviation Administration, 2024). Rashidi (2026) argues that AI is the first technology able to automate the entry-level cognitive work through which professional judgment forms. CivicWorkOS does not claim to have discovered skill erosion; it makes the erosion a quantity the allocator estimates and constrains.

2.4 Control Transfer and Competence-Aware Autonomy

Automated driving has the most developed account of control transfer. Takeover time depends on time budget, non-driving task, and the driver's prior engagement (Gold et al., 2013), and a meta-analysis of 129 studies quantifies these determinants (Zhang et al., 2019); when no takeover occurs, the system must reach a minimal-risk condition (SAE International, 2021). On the machine side, competence-aware systems learn their own competence at each level of autonomy and request human help where it is low (Basich et al., 2020), and robots can assess and communicate their proficiency to human teammates (Norton et al., 2022). Safe human–robot interaction is surveyed by Lasota et al. (2017), and the ISO 10218 series sets hardware-level requirements (International Organization for Standardization, 2025a,b). These works ask whether the *machine* is competent; takeover also needs the *human* to be competent, and that competence is what an efficiency-first allocation silently spends.

2.5 Reproducible Evaluation

Learning-based systems are notoriously sensitive to seeds, implementation details, and reporting choices (Henderson et al., 2018), and reproducibility programs ask for released code, fixed seeds, and pre-declared metrics (Pineau et al., 2021). Human–robot collaboration metrics such as fluency (Hoffman, 2019) are well developed for a single team, but no shared protocol evaluates allocation policies on long-horizon human capability. Section 5 contributes one.

2.6 Gap and Research Questions

Table 1 positions CivicWorkOS against the closest frameworks. The gap is one of integration: each component has a precedent, but no allocation mechanism we know of learns participant capabilities, models proficiency as a stock that practice builds and gaps erode, preserves it under workload limits, conditions takeover on human readiness, and comes with a reproducible protocol that tests all of this on the same task streams.

Table 1. Positioning of CivicWorkOS against the closest allocation and adaptive-automation frameworks, as the authors read each work. Columns: task–agent fit; capability estimates learned from data; human proficiency modeled as changing over time; operator workload; practice deliberately preserved; control transfer conditioned on human readiness; reproducible protocol with released code. ✓ present and specified; ▲ partially present; ✗ absent.

Framework	Fit	Learned	Proficiency dynamics	Workload	Practice preserved	Ready takeover	Repro-ducible
Ranz et al. (2017)	✓	✗	✗	✗	✗	✗	✗
Merlo et al. (2023)	✓	▲	✗	▲	✗	✗	✗
Gombolay et al. (2015)	✓	✗	✗	▲	✗	✗	✗
Chen et al. (2018)	✓	✓	✗	✗	✗	✗	▲
Kaber and Endsley (2004)	✗	✗	✗	✓	▲	▲	✗
Aricò et al. (2016)	✗	✓	✗	✓	✗	✗	✗
Basich et al. (2020)	✗	✓	✗	✗	✗	▲	▲
Nurse rostering (Burke et al., 2004)	✓	✗	▲	✓	▲	✗	✗
CivicWorkOS	✓	✓	✓	✓	✓	✓	✓

Five research questions follow.

RQ1 How can tasks be allocated among humans, AI agents, robots, and hybrid teams so that service quality, cost, safety, workload, and long-term human capability are weighed jointly?

RQ2 How can an allocator learn task requirements and participant capabilities from outcomes and multimodal observations, and use the uncertainty of those estimates?

RQ3 How much practice must be protected to form and retain human capability, when is that volume attainable, and what does it cost?

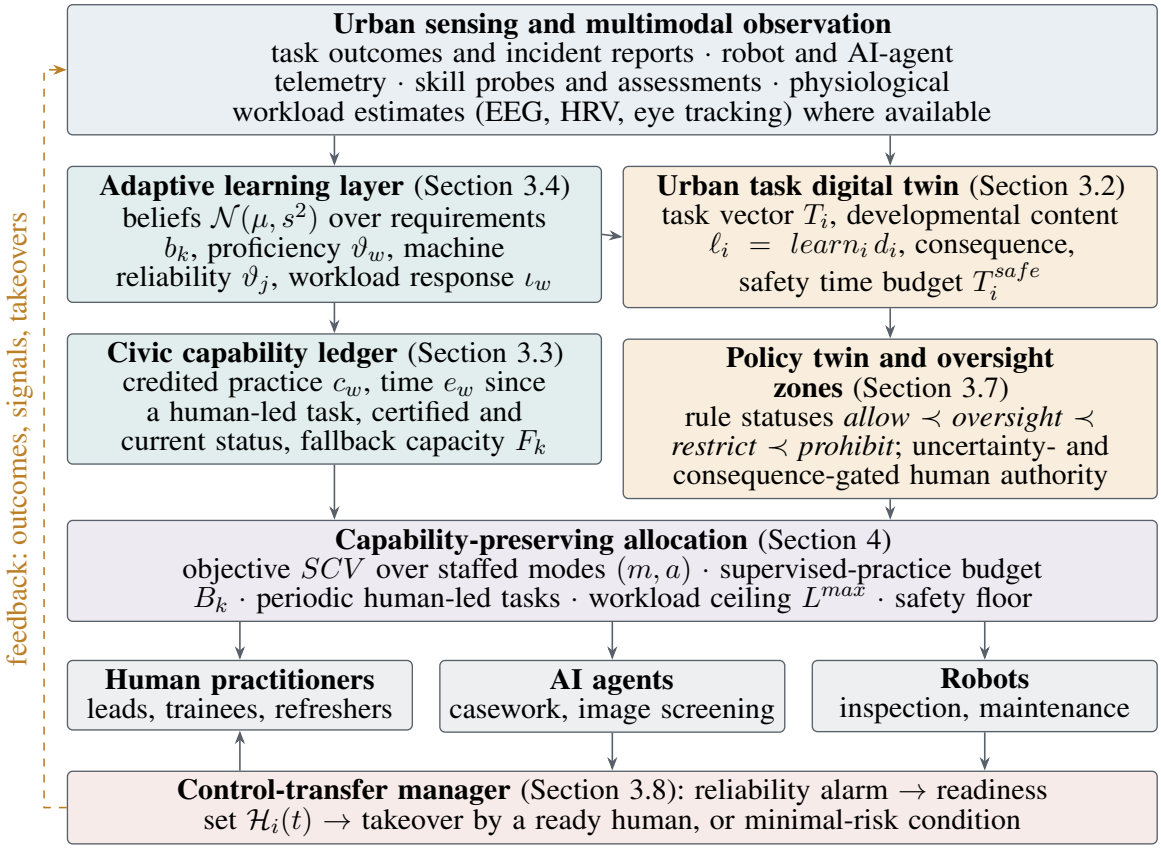


Figure 1. Reference architecture of CivicWorkOS. Solid arrows are the forward allocation path; the dashed arrow is the feedback path through which task outcomes, multimodal signals, and takeover results update the beliefs of the adaptive learning layer. The learning layer also refines the task twin's requirement estimates. The control-transfer manager acts on AI agents and robots at execution time and returns control to a human practitioner drawn from the readiness set.

RQ4 How should control be transferred when an AI agent or robot becomes unreliable, given that human readiness is itself a product of past allocation?

RQ5 How can capability-preserving allocation be compared with static and efficiency-focused allocation reproducibly, across routine operations, demand surges, and automation disruptions?

3 THE CIVICWORKOS FRAMEWORK

3.1 Architecture

Figure 1 shows the layers. Multimodal observations and task outcomes feed an adaptive learning layer that maintains beliefs over every latent quantity the allocator needs. A capability ledger records each practitioner's credited practice and recency, a policy twin turns rules into machine-checkable constraints and oversight zones, and the allocation layer chooses staffed modes subject to the capability-preservation constraints. A control-transfer manager watches execution and hands control of a failing machine to a ready human. Every outcome returns to the learning layer, closing the loop.

3.2 Tasks, Participants, and Staffed Modes

A city has urban tasks $\mathcal{T} = \{T_1, \ldots, T_n\}$, practitioners $w \in \mathcal{W}_k$ qualified in capability domain k, and AI agents and robots $j \in \mathcal{J}$. Each task belongs to a domain $k(i)$ and a critical service $\sigma(i)$. The execution

modes are given in Equation (1):

$$\mathcal{M} = \{H,\ A,\ R,\ H{+}A,\ H{+}R,\ A{+}R,\ H{+}A{+}R\}. \tag{1}$$

Each task carries a requirement vector and a duration, Equation (2):

$$T_i = \{cog_i, phy_i, emp_i, risk_i, auth_i, priv_i, urg_i, learn_i, crit_i;\ d_i\}, \tag{2}$$

covering cognitive, physical, and social demand, safety risk, legal-authority requirement, privacy sensitivity, urgency, human learning value, and criticality, each normalized to $[0, 1]$, and the execution time d_i in hours. Its *developmental content*, Equation (3),

$$\ell_i = learn_i \cdot d_i \quad \text{[qualified-practice hours]}, \tag{3}$$

is the unit in which practice is counted. The task twin also records a latent *requirement level* b_i, the proficiency a performer needs for a likely success, with domain mean b_k, and a *safety time budget* T_i^{safe}, the longest interval within which control of the task can be safely taken over.

A mode m has $n_m \geq 0$ human slots. A *roster* a fills them with distinct available practitioners and a work split $\omega_{a,w} \geq 0$, $\sum_{w \in a} \omega_{a,w} = 1$. The pair (m, a) is a *staffed mode*, and the decision variable is Equation (4):

$$x_{i,m,a} \in \{0, 1\}, \qquad \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} = 1 \quad \forall i, \tag{4}$$

where $\mathcal{A}_{i,m}$ is the set of feasible rosters. Selecting a mode alone cannot say who practices; the roster can. The *human developmental share* $\phi_m \in [0, 1]$ of a mode is the share of the task's judgment a human still exercises, Equation (5):

$$\phi_H = 1, \qquad \phi_A = \phi_R = \phi_{A+R} = 0, \tag{5}$$

with hybrid values assessed per domain. Collaborative execution, a mode with $0 < \phi_m < 1$, therefore keeps a human in the loop at reduced practice.

3.3 Human Proficiency as a Changing Resource

Each practitioner accumulates *credited practice* $c_w(t)$, the qualified-practice hours earned on assigned tasks: a task in staffed mode (m, a) credits $\ell_i\, \phi_m\, \omega_{a,w}$ to each roster member. Credited practice supports an *attainable proficiency*, Equation (6),

$$\vartheta_w^{\infty}(t) = 1 - (1 - \vartheta_0) \exp\!\big(-c_w(t)/c_k^0\big), \qquad c_k^0 = \frac{h_k}{\ln\big[(1 - \vartheta_0)/(1 - \vartheta^{comp})\big]}, \tag{6}$$

a saturating learning curve from novice proficiency ϑ_0, with c_k^0 chosen so that $\vartheta_w^{\infty} = \vartheta^{comp}$ exactly when $c_w = h_k$, the credited hours to competence. Current proficiency decays with the time $e_w(t)$ since the practitioner's last *human-led* task, one in which $\phi_m\, \omega_{a,w} \geq \varphi^{led}$, toward a residual level ϑ^{res}, Equation (7):

$$\vartheta_w(t) = \vartheta^{res} + \big(\vartheta_w^{\infty}(t) - \vartheta^{res}\big) \exp\!\big(-e_w(t)/\tau_w^{ret}\big), \tag{7}$$

with retention time constant τ_w^{ret}; a human-led task resets e_w to zero, the alternation of learning and forgetting that learn–forget curve models describe (Jaber and Bonney, 1996). Equation (7) applies while $\vartheta_w^{\infty} > \vartheta^{res}$.

This separates two statuses that municipal records conflate. A practitioner is *certified* when $c_w \geq h_k$, which is the records-based career stage of Equation (8),

$$\mathrm{stage}_k(w, t) = \begin{cases} \textit{developing}, & c_w(t) < h_k, \\ \textit{certified}, & c_w(t) \geq h_k, \end{cases} \tag{8}$$

and *current* when $\vartheta_w(t) \geq \vartheta^{comp}$. Only a current practitioner can be relied on to supervise, decide, or take over. The *developmental eligibility* of a roster, the share of its human work performed by developing practitioners, is Equation (9):

$$\psi_a(t) = \sum_{w \in a} \omega_{a,w} \mathbb{1}\Big[\mathrm{stage}_{k(i)}(w,t) = \textit{developing}\Big] \in [0,1]. \tag{9}$$

PROPOSITION 1 (Recency interval). *Under Equation* (7)*, a certified practitioner with attainable proficiency $\vartheta_w^{\infty} \geq \vartheta^{comp} > \vartheta^{res}$ at the start of a gap remains current if the interval between consecutive human-led tasks does not exceed*

$$\bar{e}_w = \tau_w^{ret} \ln \frac{\vartheta_w^{\infty} - \vartheta^{res}}{\vartheta^{comp} - \vartheta^{res}}. \tag{10}$$

The condition is also necessary when no practice is credited between human-led tasks. Keeping N_k certified practitioners current over a period ΔT then requires at least $N_k \lceil \Delta T / \bar{e} \rceil$ human-led tasks, where $\bar{e}$ is the smallest interval among them.

Proposition 1 turns decay into a schedule. A practitioner whose record barely meets certification ($\vartheta_w^{\infty} = \vartheta^{comp}$) has $\bar{e}_w = 0$: certification without consolidation leaves no margin, which is why newly certified practitioners need continued human-led work. The required volume is small in hours but must be scheduled, because it cannot be banked: practice beyond the interval does not extend it. Aviation's recency rules encode the same logic (Federal Aviation Administration, 2024). Certification schemes count raw clock hours h_k^{raw} of unshared, unassisted practice, which accrue the mean learning value per clock hour, so $h_k = \overline{learn}_k \, h_k^{raw}$ (Equation (11)),

$$h_k \;=\; \overline{learn}_k \cdot h_k^{raw}, \tag{11}$$

and a practitioner working machine-assisted or beside a lead accrues less per clock hour (Section 6.1).

The *capability ledger* holds, per domain, c_w and e_w for every practitioner, the counts $N_k(t)$ of certified and $N_k^{dev}(t)$ of developing practitioners, and the *fallback capacity*, Equation (12),

$$F_k(t) = \sum_{w \in \mathcal{W}_k} \zeta_w(t) \mathbb{1}\Big[\hat{\vartheta}_w(t) - z\, s_w(t) \geq \vartheta^{comp}\Big], \tag{12}$$

the surge-deployable hours $\zeta_w(t)$ of practitioners whose learned proficiency (Section 3.4) is current with confidence z. Counting certified rather than current practitioners would overstate fallback precisely when practice has lapsed.

3.4 Adaptive Learning Mechanism

The allocator never observes proficiency, requirement, or reliability directly. It maintains independent Gaussian beliefs $\mathcal{N}(\mu, s^2)$ over every latent quantity: a proficiency offset o_w for each practitioner, so that $\vartheta_w = \vartheta_w^{nom} + o_w$ where ϑ_w^{nom} is Equation (7) evaluated at the logged c_w and e_w with nominal τ^{ret}; a reliability ϑ_j for each AI-agent pool or robot fleet; a requirement level b_k for each domain; and a workload response ι_w for each practitioner. The offset absorbs everything the practice log cannot see, such as individual learning and forgetting rates.

Outcome model. A staffed mode's success probability follows a logistic link, Equation (13):

$$\Pr(y_i = 1 \mid m, a) = \sigma\big(\beta_0\, y^*_{i,m,a}\big), \qquad y^*_{i,m,a} = \phi_m\, \bar{\vartheta}_a + (1 - \phi_m)\, \vartheta_{j(m)} - b_i, \tag{13}$$

where $\sigma(\cdot)$ is the logistic function (not to be confused with the service map $\sigma(i)$), β_0 its slope, $y^*_{i,m,a}$ the linear predictor, $\vartheta_{j(m)}$ the reliability of the machine component of mode m, and $\bar{\vartheta}_a$ the team proficiency of roster a. For a lead w_{ld} supervising a second practitioner w_{sp} (a trainee or a refresher) with work share ω, $\bar{\vartheta}_a = \vartheta_{w_{\mathrm{ld}}} - \omega(1 - \kappa^{sup})(\vartheta_{w_{\mathrm{ld}}} - \vartheta_{w_{\mathrm{sp}}})$, where the supervision recovery κ^{sup} is the share of the proficiency

gap that the lead's oversight recovers. Equation (13) is the item-response form of learner models (Pelánek, 2017), extended to teams of humans and machines.

Outcome update. Because y^* is linear in the latent quantities, an observed outcome updates all of them at once by assumed-density filtering (Minka, 2001): the logistic likelihood is replaced by its probit approximation $\Phi(\varkappa\beta_0 y^*)$, with Φ the standard normal distribution function and $\varkappa = \sqrt{\pi/8}$, and the posterior is projected onto independent Gaussians by moment matching. Writing $y^* = \sum_q \alpha_q x_q$ with belief means μ_q and variances s_q^2, and setting $\bar{y}^* = \sum_q \alpha_q \mu_q$, $v = \sum_q \alpha_q^2 s_q^2$, $\kappa = \varkappa\beta_0$, and $t = (2y-1)\kappa\bar{y}^*/\sqrt{1+\kappa^2 v}$, Equation (14) gives the update:

$$\mu_q \leftarrow \mu_q + (2y-1)\frac{\alpha_q s_q^2\,\kappa\,\nu(t)}{\sqrt{1+\kappa^2 v}}, \qquad s_q^2 \leftarrow s_q^2\Big[1 - \frac{\alpha_q^2 s_q^2\,\kappa^2\,\nu(t)\big(\nu(t)+t\big)}{1+\kappa^2 v}\Big], \tag{14}$$

with $\nu(t) = \Phi'(t)/\Phi(t)$ the inverse Mills ratio. Credit for a failure is thus shared among the lead, the trainee, the machine, and the requirement in proportion to each one's coefficient and current uncertainty, the same update used by dynamic rating systems (Glickman, 1999).

Multimodal observations. Where a direct but noisy observation $o_q^{(n)} = x_q + \epsilon^{(n)}$, $\epsilon^{(n)} \sim \mathcal{N}(0, \sigma_n^2)$, of a latent quantity is available, it enters by the scalar Kalman update of Equation (15):

$$\mu_q \leftarrow \mu_q + \frac{s_q^2}{s_q^2+\sigma_n^2}\big(o_q^{(n)} - \mu_q\big), \qquad s_q^2 \leftarrow \frac{s_q^2\,\sigma_n^2}{s_q^2+\sigma_n^2}. \tag{15}$$

Three channels are modeled: skill probes or supervisor assessments observe ϑ_w; robot self-diagnostics and AI-agent monitoring observe ϑ_j (Norton et al., 2022); and physiological workload estimates fused from electroencephalography (EEG), heart-rate variability (HRV), or eye tracking (Debie et al., 2021; Dehais et al., 2020) observe ι_w. Each channel is optional, which is what "where available" means operationally: a missing channel leaves the belief to the outcome update, and a channel's noise variance σ_n^2 fixes its weight.

Non-stationarity. Between observations each variance grows by a process term, $s_q^2 \leftarrow s_q^2 + \varsigma_q^2$ per day, so that beliefs track drift in proficiency and machine reliability rather than freezing on old evidence. Proficiency decay itself is carried by the nominal model, so the offset's process term need only absorb individual deviations.

Workload. A practitioner's daily workload index is $L_w = \iota_w \sum_i d_i\,\varpi_m/\bar{d}$, where ϖ_m is the per-hour human demand of mode m, $\bar{d}$ the hours available to the task stream, and ι_w the practitioner's latent workload response. A takeover adds unplanned demand. Overload is $L_w > L^{max}$.

The mechanism is deliberately simple. Every belief is two numbers with a direct reading ("this inspector is estimated 0.04 below the nominal curve, give or take 0.05"), every update is a closed form costing $O(|a|+2)$ operations for a roster of $|a|$ practitioners, and the full state of a domain fits in a table an auditor can inspect. It needs no training data from other cities. Two idealizations should be explicit. The multimodal channels are modeled as unbiased observations with known noise variance; real physiological workload indices are biased, drift, and differ between people (Dehais et al., 2020; Debie et al., 2021), so a deployment would calibrate each channel per person and inflate σ_n^2 accordingly. And the outcome model is a single logistic link per domain; richer learners can replace it without changing the allocation rule, which consumes only means and variances. Beliefs about individuals are used only inside the allocator; reported quantities are aggregates (Section 7.6).

3.5 Objective: Sustainable Civic Value and Civic Automation Debt

For each task and staffed mode, the *Sustainable Civic Value* (SCV) is given by Equation (16):

$$\begin{aligned} SCV_{i,m,a} = {} & w_1 Q_{i,m,a} + w_2 S_{i,m,a} + w_3 P_{i,m,a} + w_4 Eq^{srv}_{i,m,a} + w_5 Tr_{i,m,a} \\ & - w_6 Cost_{i,m,a} - w_7 En_{i,m} - w_8 Pr_{i,m} - w_9 CAD_{i,m,a}, \end{aligned} \tag{16}$$

with service quality Q, safety S, productivity P, equity of service to residents Eq^{srv}, trust Tr, operating cost $Cost$, energy En, privacy risk Pr, and *Civic Automation Debt* (CAD), Equation (17),

$$CAD_{i,m,a} = \alpha\, D^{skill}_{i,m,a} + \beta\, D^{fall}_{i,m} + \gamma\, D^{acct}_{i,m} + \delta\, D^{dep}_{i,m} + \epsilon\, D^{trans}_{i,m,a}, \tag{17}$$

a per-task penalty for five deferred liabilities: skill-formation loss, fallback erosion, accountability dilution, vendor dependency, and labor-transition burden. The first eight terms are flows on the task's execution window; the debt components are marginal proxies for changes in municipal stocks, so the two are measured in different domains and do not double-count. Skill-formation loss is the component that carries the framework, Equation (18):

$$D^{skill}_{i,m,a} = 1 - \phi_m\, \psi_a, \tag{18}$$

since a task performed entirely by machine forms nobody, and neither does one performed by an already-certified practitioner alone. Appendix S3.1 defines the other four. In the online rule, Q, S, and P are evaluated on the learned success probability of Equation (13), so the objective moves as the beliefs do.

All terms are rescaled to $[0, 1]$. The illustrative weights are w_1=0.15 (quality), w_2=0.18 (safety), w_3=0.14 (productivity), w_4=0.10 (service equity), w_5=0.05 (trust), w_6=0.08 (cost), w_7=0.03 (energy), w_8=0.05 (privacy), and w_9=0.22 (debt), with debt weights α=0.28, β=0.22, γ=0.18, δ=0.12, and ϵ=0.20. They are design defaults for a municipal review panel to revise, not fitted values. Composed, the deferred block carries 0.22 of the weight, but skill formation alone carries only $w_9\alpha = 0.062$, sixth among the constructs, behind safety at 0.180. A weight can be outvoted; this is why capability preservation is carried mainly by constraints (Proposition 4).

3.6 Capability-Preservation Constraints

Four mechanisms preserve capability. Three are constraints rather than weights; the fourth, collaborative execution, acts through the practice credit that those constraints count.

Supervised practice. The *Human Capability Preservation Budget* reserves a volume of developmental practice delivered to practitioners who are still developing, Equation (19):

$$\Lambda_k(\Delta T) \;=\; \sum_{i \in \mathcal{T}_k(\Delta T)} \sum_{m} \sum_{a} x_{i,m,a}\, \ell_i\, \phi_m\, \psi_a \;\geq\; B_k(t), \qquad B_k(t) = \max\Big(B^{min}_k,\; \eta_k\, N_k(t)\, r_k(t)\, h_k\Big), \tag{19}$$

where r_k is the annual attrition of certified practitioners, η_k a replacement factor set by policy, and B^{min}_k a floor. The factor ψ_a restricts credit to developing practitioners: without it, a domain that gives every task to its incumbents meets a budget sized from novice formation while forming nobody. The budget constrains a volume of practice, not a number of posts, so a task may go to a developing practitioner supported by an AI agent even where the AI agent could perform it alone.

Collaborative execution. Hybrid modes with $0 < \phi_m < 1$ earn proportionally reduced credit, so the allocator can trade machine assistance against practice explicitly rather than by default.

Periodic human-led tasks. For every certified practitioner, a human-led task must occur within the recency interval of Proposition 1, Equation (20):

$$\sum_{i:\, t - \bar{e}_w \leq t_i \leq t} \sum_{m,a} x_{i,m,a}\, \mathbb{1}\big[\, \phi_m\, \omega_{a,w} \geq \varphi^{led} \,\big] \;\geq\; 1 \qquad \forall\, w:\ \mathrm{stage}_k(w,t) = \textit{certified},\ \forall t, \tag{20}$$

enforced online with a guard band: a practitioner becomes *due* once e_w reaches a fraction g^{ret} of the interval. The interval is computed from the logged practice record, Equation (10) evaluated at $\vartheta^{\infty}_w(c_w)$ with nominal τ^{ret}, so the schedule is auditable from records; the learned estimate of Section 3.4 decides instead whether a practitioner is *current*, and so whether they may lead or take over. A practitioner whose learned estimate has fallen below ϑ^{comp} is restored by a *supervised refresher*, a human-led task beside a current lead.

Workload ceiling. No assignment may push a practitioner's estimated workload above a planning margin ϱ^L of the ceiling, Equation (21):

$$\hat{\iota}_w \sum_{i \in \mathcal{T}(t)} \sum_{m,a} x_{i,m,a} \mathbb{1}[w \in a]\, d_i\, \varpi_m \;\le\; \varrho^L\, L^{max}\, \bar{d} \qquad \forall w,\ \forall t, \tag{21}$$

where $\mathcal{T}(t)$ is the day's tasks and $\hat{\iota}_w$ the learned workload response, so that practitioners whose physiological signals show high strain are planned with less load.

Two distributional constraints complete the set. The *Capability Access Constraint* requires each worker group g to receive at least a published target share $\theta_{k,g} - \varepsilon_k$ of protected practice, so that a budget sized from the incumbent workforce does not reproduce its composition by default, and the *Just Transition Constraint* caps the fractional loss of allocated hours of any payroll group at τ_g per period. Appendix S4 states both and the legal analysis they require. Finally, the *resilience reserve* requires every critical service to keep, after losing its n_s largest automation components, a surviving capacity of at least $R_s^{min} = \kappa_s D_s^{peak}$, Equation (22):

$$Res_s^{(n_s)}(t) \;=\; \sum_{c \in \mathcal{C}_s^{aut} \setminus \mathcal{C}_s^{(ns)}} C_{s,c}(t) \;+\; \sum_{k \in \mathcal{K}_s} \xi_{s,k} F_k(t) \;\ge\; \kappa_s\, D_s^{peak}, \tag{22}$$

the N−1/N−2 criterion of reliability engineering (Jain et al., 2024; Ouyang, 2014) with the human term counted through the readiness-weighted F_k of Equation (12).

3.7 Policy Twin and Uncertainty-Gated Oversight

A policy digital twin encodes laws, collective agreements, and local rules as machine-checkable rules, each returning a status in the lattice *allow* $\prec$ *allow-with-oversight* $\prec$ *restrict* $\prec$ *prohibit*; the status of a staffed mode is the join over the rules that fire, so the most restrictive applies (Appendix S3.3). Task classes are placed in the four oversight zones of Table 2.

Table 2. Adaptive oversight zones and the staffed modes each admits. Zone membership is evidence-conditioned: a class is demoted toward Z4 when incidents, upheld appeals, or rising debt are observed, or when the learned reliability of its automation falls, and promoted only by an affirmative review-panel decision.

Zone	Allocation principle	Admitted staffed modes
Z1 Autonomous	High-confidence, low-consequence tasks may be performed autonomously.	All $m \in \mathcal{M}$, subject to the uncertainty gate of Equation (23).
Z2 Augmented	AI agents or robots perform substantial work while a human remains involved.	m with $\phi_m > 0$; any feasible roster.
Z3 Human authority	AI may advise, but a designated human makes or validates the decision.	m with $\phi_m \ge 0.5$; roster names a current decision-maker of record.
Z4 Human reserved	Automation restricted for reasons of dignity, democratic authority, capability preservation, or unacceptable risk.	$m \in \{H,\ H{+}R\}$, the robot under direct control; roster contains a current practitioner.

Learning makes oversight adaptive at the level of the single task. A machine-only mode is admissible only if the decision is not consequential and its predicted success is not too uncertain, Equation (23):

$$crit_i \cdot auth_i < q^{Z3} \qquad \text{and} \qquad \mathrm{sd}\big[\hat{p}_{i,m,a}\big] \approx \hat{p}(1-\hat{p})\,\beta_0\sqrt{v_{i,m,a}} \;\le\; u^{max}, \tag{23}$$

where $\hat{p}$ is the posterior-predictive success probability and v the variance of the linear predictor of Equation (13). A task failing either condition is routed to a mode with human authority. Where the policy twin already classifies decisions as consequential, as for benefit determinations in casework, that classification replaces the product $crit_i\, auth_i$ in the first condition. The gate opens as evidence accumulates about a machine and closes again when telemetry or failures make its reliability uncertain, so human oversight is spent where the estimates are weakest (Basich et al., 2020).

3.8 Readiness-Constrained Control Transfer

When an AI agent or robot fails during a task, or its learned reliability triggers an alarm, control must pass to a human. In a collaborative mode the human in the loop takes over directly. In a machine-only mode a practitioner must be found, and the right one is not simply the most senior or the least busy. CivicWorkOS restricts the choice to the *readiness set*, Equation (24):

$$\mathcal{H}_i(t) = \Big\{\, w \in \mathcal{W}_{k(i)} :\; \hat{\vartheta}_w - z\, s_w \ge \vartheta^{comp}, \quad \hat{L}_w + \Delta L_i^{to} \le \varrho^{to} L^{max}, \quad \hat{T}_w^{to}(i) \le T_i^{safe} \,\Big\}, \tag{24}$$

practitioners who are current with confidence, have workload headroom for the takeover's demand ΔL_i^{to}, and are expected to take over within the task's safety time budget. The expected takeover time, Equation (25),

$$\hat{T}_w^{to}(i) = T^{det} + T^{reach} + T^{sa}\,\big[1 + \chi^{\vartheta}\max(0, \vartheta^{hi} - \hat{\vartheta}_w)\big]\big[1 + \chi^{ootl}\,\mathbb{1}(e_w > e^{ootl})\big]\big[\chi^{L}\big]^{\mathbb{1}(\hat{L}_w > L^{hi})}, \tag{25}$$

sums detection, reach, and situation-awareness time, and lengthens the last for lower proficiency, for being out of the loop (Endsley and Kiris, 1995; Gold et al., 2013), and for high workload, the determinants identified for automated driving (Zhang et al., 2019). The takeover goes to the member of $\mathcal{H}_i(t)$ with the highest estimated success, and if $\mathcal{H}_i(t) = \emptyset$ the machine is commanded to a minimal-risk condition, such as a safe stop with the task re-queued (SAE International, 2021), rather than handed to an unready human. The rule makes readiness a constraint, and readiness is itself produced by past allocation: Equation (20) is what keeps $\mathcal{H}_i(t)$ non-empty.

4 THE ALLOCATION PROGRAM AND ONLINE RULE

4.1 The City-Wide Program

The constraints couple decisions across tasks, so over a rebalancing window CivicWorkOS solves program (26):

$$\max_{\mathbf{x}} \quad \sum_i \sum_m \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a}\, SCV_{i,m,a} \tag{26a}$$

$$\text{s.t.} \quad \textstyle\sum_m \sum_a x_{i,m,a} = 1, \qquad \forall i, \tag{26b}$$

$$x_{i,m,a} = 0 \;\text{ if } \mathcal{P}(T_i, m, a, t) = \textit{prohibit} \text{ or Equation (23) fails}, \tag{26c}$$

$$x_{i,m,a} = 0 \;\text{ if } \hat{p}^{low}_{i,m,a} < S_i^{min}, \tag{26d}$$

$$\text{Equations (19), (20), and (21)}, \qquad \forall k, w, \tag{26e}$$

$$\text{access, just-transition, and reserve constraints (Appendix S4, Equation (22))}, \qquad \forall k, g, s, \tag{26f}$$

$$\textstyle\sum_i \sum_m \sum_a x_{i,m,a}\, u_{i,m,a,r} \le U_r, \qquad \forall r, \tag{26g}$$

$$x_{i,m,a} \in \{0, 1\}, \tag{26h}$$

where the safety floor S_i^{min} is applied to the lower confidence bound $\hat{p}^{low}$ of the learned success probability, so that uncertainty tightens rather than loosens safety, and $u_{i,m,a,r}$ is the consumption of resource r (human hours by stage, fleet hours, inference budget) against availability U_r. The program is a binary multidimensional assignment problem with side constraints, NP-hard in general, and is solved periodically; this article makes no scalability claim for it. The safety floor is set on the scale of the quantity it bounds: the worked allocation of Section 6.1, whose estimates are fixed, bounds the normalized safety term S by $risk_i\, crit_i$, whereas the testbed of Section 5 bounds predicted success by $0.55 + 0.35\, risk_i\, crit_i$ (Appendix S1).

4.2 Online Rule and Admissibility

Between rebalances, tasks are decided on arrival. CivicWorkOS relaxes the coupling constraints with dual multipliers from the most recent solution, so the online score is Equation (27):

$$\begin{aligned}\widetilde{SCV}_{i,m,a} = SCV_{i,m,a} + \lambda_{k(i)}\, \ell_i\, \phi_m\, \psi_a + \mu_{\sigma(i)}\, \Delta^{res}_{\sigma(i)}(i,m,a) \\ - \sum_r \upsilon_r\, u_{i,m,a,r} + (\text{access and transition terms}),\end{aligned} \tag{27}$$

with λ_k the capability dual and Δ^{res} the task's counterfactual contribution to the reserve (Appendix S2.2). The budget is a state of the domain, not of a mode, so it is enforced through a catch-up test: a staffed mode is admissible for the budget if the period's accrued practice $\Lambda_k(t)$ is on its pro-rated trajectory $\bar{B}_k(t) = B_k\, t/\Delta T$, or if it carries at least the per-task share of the remaining gap, Equation (28):

$$\Lambda_k(t) \geq \bar{B}_k(t) \quad \text{or} \quad \ell_i\, \phi_m\, \psi_a \;\geq\; \delta_k(t) = \frac{\max\big(0,\; B_k(t) - \Lambda_k(t)\big)}{n_k^{rem}(t)}, \tag{28}$$

with n_k^{rem} the tasks still expected in the period; the test tightens as the gap grows. Retention is enforced the same way: when a practitioner is due, the admissible set is first restricted to rosters that give them a human-led task or that restore a lapsed practitioner. Algorithm 1 states the procedure and Figure 2 its control flow. When the restricted sets are empty the rule falls back to the next stage rather than leaving the task unassigned, and records the fallback.

Algorithm 1 Online capability-preserving allocation with adaptive learning and control transfer

Require: Task T_i; time t; beliefs $\{\mu, s^2\}$; ledger $\{c_w, e_w\}$; policy twin $\mathcal{P}$; budgets B_k; duals
Ensure: Selected staffed mode $(m,a)_i^*$, outcome, updated beliefs

1: $k \leftarrow k(i)$; due set $\mathcal{D} \leftarrow \{w$ certified : $e_w \geq g^{ret}\, \bar{e}_w,\ \hat{\vartheta}_w \geq \vartheta^{comp}\}$; $\delta_k \leftarrow$ (28)
2: **for** each $m \in \mathcal{M}$ with $\mathcal{P}(T_i, m, t) \neq$ *prohibit* and machine capacity available **do**
3: rosters: best-estimated current lead; most-due lead in $\mathcal{D}$; each with and without a trainee; a lapsed practitioner as refresher beside a current lead
4: keep rosters satisfying the workload ceiling (21)
5: **for** each roster a **do**
6: $\hat{p}$, $\hat{p}^{low} \leftarrow$ Equation (13) on current beliefs; discard if $\hat{p}^{low} < S_i^{min}$ or gate (23) fails
7: $\widetilde{SCV}_{i,m,a} \leftarrow$ (27)
8: **end for**
9: **end for**
10: stages: (retention: due lead or refresher) $\rightarrow$ (budget catch-up (28)) $\rightarrow$ (all admissible)
11: $(m,a)_i^* \leftarrow \arg\max \widetilde{SCV}$ over the first non-empty stage; if none, hold the task (Zone Z4 queue)
12: Execute. **If** the machine component faults: take over by the in-loop human, or by $\arg\max_{w \in \mathcal{H}_i(t)} \hat{\Pr}(\text{success})$ from (24); if $\mathcal{H}_i(t) = \emptyset$, command a minimal-risk condition
13: Observe y_i; update beliefs by (14); apply available multimodal observations by (15)
14: For each $w \in a^*$: $c_w \leftarrow c_w + \ell_i \phi_{m^*} \omega_{a^*,w}$; reset e_w if $\phi_{m^*} \omega_{a^*,w} \geq \varphi^{led}$; $\Lambda_k \leftarrow \Lambda_k + \ell_i \phi_{m^*} \psi_{a^*}$

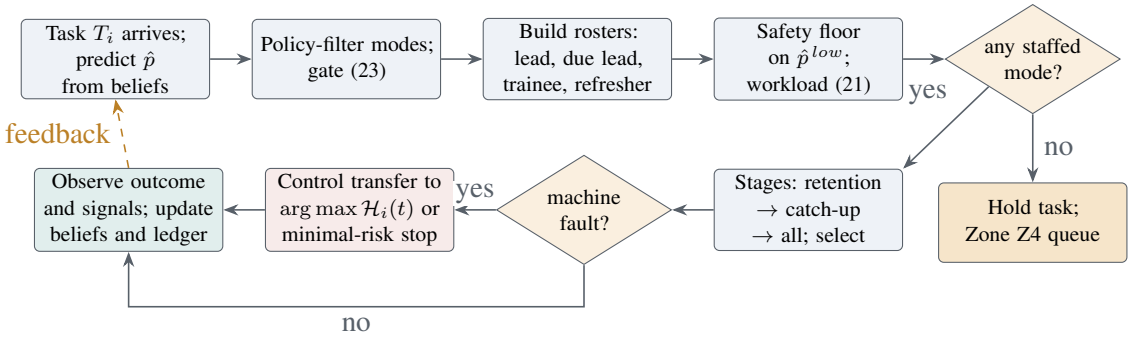


Figure 2. Runtime control flow of Algorithm 1. The upper row predicts success from the current beliefs, filters modes through the policy twin and the uncertainty gate, constructs rosters, and applies the safety floor and the workload ceiling. The lower row selects through the retention and budget stages, executes, transfers control on a machine fault, and feeds the outcome and any multimodal signals back into the beliefs (dashed).

Per task the rule evaluates at most seven modes and a handful of rosters each, so it is linear in arrivals. It is exact only when the duals are exact and the state has not drifted.

4.3 Formal Properties

Proofs are in Appendix S2. A budget the task stream cannot supply is not demanding but infeasible.

PROPOSITION 2 (Feasibility of the capability budget). *Let* $\Phi_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \ell_i$ *and let* $\bar{\zeta}_k$ *be the largest credited share* $\phi_m \psi_a$ *of any staffed mode passing the policy and safety conditions for some task of domain* k*. Then Equation* (19) *is satisfiable only if* $B_k(t) \le \Phi_k(\Delta T)\,\bar{\zeta}_k$*. Where* B_k *is attained by its second argument, this is a ceiling on attrition, Equation* (29)*:*

$$r_k(t) \;\le\; r_k^{crit} \;=\; \frac{\Phi_k(\Delta T)\,\bar{\zeta}_k}{\eta_k\, N_k(t)\, h_k}. \tag{29}$$

When the condition fails, the framework flags infeasibility, publishes the shortfall, and reports the restoring actions: relax the supervision ratio where risk permits, procure less autonomous tooling, expand the supervised task stream, or lower η_k as a recorded decision to let the domain shrink.

PROPOSITION 3 (Required trainee cohort). *Let* Θ_k *be a developing practitioner's domain clock hours per period, and* $\bar{\phi}_k$ *and* $\bar{\omega}_k$ *the duration-weighted means of* ϕ_m *and of the practitioner's work share* $\omega_{a,w}$ *over those hours. If the mean of* $learn_i\, \phi_m\, \omega_{a,w}$ *factorizes (as under mutual independence), the expected time to certification is Equation* (30)*,*

$$\tau_k \;=\; \frac{h_k^{raw}}{\Theta_k} \cdot \frac{1}{\bar{\phi}_k\, \bar{\omega}_k}, \tag{30}$$

and the domain must hold a standing cohort of $\hat{n}_k = \eta_k\, N_k\, r_k\, \tau_k$ *developing practitioners (Equation* (31)*),*

$$\hat{n}_k \;=\; \eta_k\, N_k(t)\, r_k(t)\, \tau_k. \tag{31}$$

Equation (30) factorizes formation time into the time certification hours imply and a *dilution factor* $1/(\bar{\phi}_k \bar{\omega}_k)$, due to machine assistance and to sharing the work with a lead.

PROPOSITION 4 (Price and constraint separation). *For task* i *let* $\Xi(w_9)$ *be the arg max of Equation* (16) *over its admissible set with* $w_1, \ldots, w_8$ *in fixed ratio, and let* $w_9^\star(i) = \inf\{w_9 \ge 0 : \Xi(w_9) \ne \Xi(0)\}$*. For every* $w_9 < w_9^\star(i)$ *the selected staffed mode is the one an objective ignoring Civic Automation Debt would select.*

Proposition 4 is a diagnostic: when $w_9 < w_9^\star$, any departure from the debt-blind choice comes from the constraints. Reporting $w_9^\star$ with each allocation shows a reviewer whether the weight or a constraint decided it.

PROPOSITION 5 (Between-rebalance violation bound). *If each task's outcome is observed before the next is decided and the admissibility test is evaluated on the post-execution state, the resource constraints* (26g) *hold exactly for admitted tasks, and the just-transition constraint is exceeded by at most the largest single upward revision $\bar{\delta}_g$ of a task's estimated displacement, independently of the number of arrivals.*

The bound is an accounting result, not a regret guarantee; Appendix S2.3 states what a regret bound for the online rule would require, and why the covering form of the budget and the endogenous coefficients of ψ_a make it non-standard.

5 EVALUATION PROTOCOL

The protocol is specified so that another group can run it unchanged, and executed once (Section 6.2). Appendix S1 gives every parameter.

5.1 Testbed

A day-stepped synthetic municipal testbed runs for three years of 250 working days. Three service domains (Table 3) span the range of human–machine interaction a city faces: robot-assisted structural inspection, where every mode keeps a human and the policy twin forbids autonomous determinations; street maintenance and sanitation with robot fleets; and citizen-service casework with AI agents. Tasks arrive as Poisson streams with per-task requirement levels drawn around the domain mean, and each practitioner has a latent learning-rate multiplier, retention time constant, and workload response, so the nominal model the allocator starts from is systematically wrong for individuals. Machine reliability follows a mean-reverting random walk. Certified practitioners leave at the domain's attrition rate, and a developing practitioner who reaches h_k is certified and replaced in the trainee post. Outcomes follow Equation (13) with the true latent values; incidents follow failures with probability proportional to task risk.

Table 3. Service domains of the reference testbed. All values are illustrative modeling choices, not calibrated estimates of any city. "Consequential" is the share of tasks whose decision carries legal or rights consequences. Reliability is the nominal reliability of the domain's robot fleet or AI-agent pool.

	Structural inspection	**Maintenance and sanitation**	**Citizen-service casework**
Machine	robot fleet (4 units)	robot fleet (7 units)	AI-agent pool (12 units)
Modes	H, $H{+}R$, $H{+}A{+}R$	H, $H{+}R$, R	H, $H{+}A$, A
Arrivals per day; duration (h)	6; 4.0	20; 2.0	40; 1.0
$learn_i$; $risk_i$; $crit_i$	0.80; 0.70; 0.90	0.50; 0.50; 0.60	0.60; 0.30; 0.70
Consequential share	1.00	0.10	0.30
Requirement b_k (prior bias)	0.62 (-0.04)	0.55 (-0.04)	0.58 (-0.04)
Deadline (days); takeover budget (min)	5; 15	1; 20	2; 45
Certified; trainee posts; h_k (h)	10; 4; 600	12; 4; 500	10; 4; 400
Machine reliability; cost per hour	0.86; 0.50	0.82; 0.30	0.88; 0.08
Annual attrition r_k	0.12	0.12	0.12

5.2 Policies

Five policies run on identical task streams. *Static* allocation fixes one mode per domain at commissioning from prior ratings ($H{+}A{+}R$, R, and A), attaches a free trainee to human-involved work as apprenticeship custom would, assigns the least-loaded certified practitioner as lead and as takeover operator, and does not learn. *Efficiency-focused* allocation uses the same learned beliefs as CivicWorkOS and maximizes immediate expected value, the value of a successful completion net of cost and expected incident loss, with a safety floor on the posterior mean; it has no capability, retention, workload, oversight, or readiness terms, and hands takeovers to the practitioner with the highest estimated proficiency. *CivicWorkOS* adds Equations (19) through (24). Two ablations isolate mechanisms: *CivicWorkOS without learning* keeps every

constraint but plans on the nominal model and prior ratings, ignoring outcomes and multimodal signals; *CivicWorkOS without retention* drops periodic human-led tasks and refreshers. Comparing CivicWorkOS with efficiency-focused allocation isolates the capability-preserving terms, because both learn identically.

The testbed exercises the framework at reduced resolution, and three simplifications should be stated. First, both learning policies score a staffed mode by its immediate expected value, $10\,(0.5 + 0.5\,crit_i)\,\hat{p} - \text{cost} - 30 \times 0.15\,risk_i(1-\hat{p})$, a three-term instance of Equation (16) (quality through $\hat{p}$, operating cost, and safety through expected incident loss) without the debt term; capability preservation enters only through the constraints, which Proposition 4 shows is where it acts at the default weights. Second, the online rule enforces the budget by the staged catch-up test of Equation (28) rather than through a dual price, because the testbed has no periodic program to supply duals. Third, consequential decisions are a per-task flag drawn at the domain's consequential share (Table 3). Every other element of Algorithm 1, including the learning updates, the safety floor on $\hat{p}^{\,low}$, the oversight gate, the workload ceiling, the retention stage with refreshers, and the readiness-constrained takeover, is implemented as specified.

5.3 Scenarios and Metrics

Routine operations run the horizon unperturbed. A *demand surge* doubles arrivals in every domain for 30 working days from day 400. An *automation disruption* degrades every robot fleet and AI-agent pool for 20 working days from day 500: true reliability falls to 0.45, the per-task fault rate rises from 0.02 to 0.25, and machine capacity halves, silently, so the allocator learns of it only through outcomes, faults, and telemetry.

Five metric families are reported. *Task completion* is the share of tasks completed before their deadline, overall and for tasks arriving in the event window. *Service cost* is total cost per completed task (wages, machine hours, incident losses), reported as an index with efficiency-focused allocation at 100 on the same seed. *Workload* is the share of practitioner-days with true workload index above L^{max}. *Skill retention* is the share of certified practitioners who are current at the end of the horizon, with the count of certified and current practitioners. *Takeover performance* is the share of takeover requests recovered by a human, overall and in the event window, with mean takeover time. Incidents per 1,000 tasks are reported descriptively.

5.4 Statistical Plan and Pre-Specified Hypotheses

Each policy runs on 30 reporting seeds per scenario; development and debugging used a disjoint seed range (1000–1009) and no reporting seed was run before the final configuration was fixed. A seed fixes every exogenous draw, including arrivals, requirements, outcome and fault draws, sensor noise, attrition times, and machine reliability paths, so comparisons are paired by common random numbers. Outcomes are reported as means with 95% t-intervals. The eight hypotheses of Table 8 and their thresholds were fixed in `configs/protocol/hypotheses.yaml` before the evaluation code was written. Each is tested by a paired two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) with Holm–Bonferroni control (Holm, 1979) over the family of 12 tests at $\alpha = 0.05$, and effect sizes are matched-pairs rank-biserial correlations (Kerby, 2014). A directional hypothesis is supported when its adjusted p-value is below 0.05, the mean paired difference has the predicted sign, and the difference meets its threshold; the two non-inferiority hypotheses are decided by whether the 95% interval of the mean paired difference lies inside the margin.

5.5 Reproducibility

The testbed, policies, estimators, and analysis are released with the article (Data Availability Statement). A single command, `python scripts/run_protocol.py --paper`, executes the 450 runs, writes every run record, the resolved configuration, the summary, and the hypothesis verdicts, and regenerates the data behind Table 7, Table 8, and Figure 4. A separate script checks every number this article prints against those files and against the arithmetic of Section 6.1. The reference execution takes 10 to 13 minutes on a 12-core workstation with Python 3.10.11, NumPy 1.26.2, and SciPy 1.11.4.

6 RESULTS

6.1 An Exact Worked Allocation: Municipal Bridge Inspection

We first carry one allocation through the framework by exact arithmetic, with beliefs taken at their means, to show what the constraints do and what they cost. The inputs are illustrative expert estimates, not measurements, and Section 6.1 maps where each conclusion survives them.

Task, Domain, and Candidate Set

T_i is one cycle of fatigue-crack assessment on a steel girder span: $cog = 0.80$, $phy = 0.60$, $emp = 0.10$, $risk = 0.70$, $auth = 0.90$, $priv = 0.20$, $urg = 0.40$, $learn = 0.80$, $crit = 0.90$, and $d_i = 10$ hours, so $\ell_i = 8$ qualified-practice hours. A structural determination carries statutory sign-off, so the policy twin returns *restrict*, admitting only rosters with a licensed certified practitioner and removing the machine-only modes. The safety floor $S_i^{min} = risk_i\, crit_i = 0.63$ removes unaided human inspection ($S = 0.55$) and human plus AI ($S = 0.60$), leaving $H{+}R$ and $H{+}A{+}R$, each with roster a_0 (a certified lead alone, $\psi = 0$) or a_1 (a lead with one developing inspector on an equal split, $\psi = 0.5$). A supervision rule of one trainee per lead caps $\bar{\zeta}_k$ at $\phi_{H+R} \times 0.5 = 0.35$. The domain generates $n_k = 2{,}100$ inspections a year, so $\Phi_k = 2100 \times 8 = 16{,}800$ qualified-practice hours, with $N_k = 24$ certified practitioners and annual attrition $r_k = 0.12$.

Scoring the Staffed Modes

Table 4 gives the normalized estimates for the eight staffed modes. Read alone it favors a fully hybrid team led by a certified inspector alone: $H{+}A{+}R/a_0$ has the best quality, safety, and productivity, the lowest cost, and the highest SCV at 0.3426, despite the largest debt. Weighting debt does not change this: the arg max first changes at $w_9^\star = 0.313$ (Equation (32)),

$$w_9^\star = 0.313, \tag{32}$$

against a default $w_9 = 0.22$. At the default, the debt weight closes 70% of the gap between the two rosters, from 0.0241 to 0.0072 objective units, but the reversal is produced by the constraint.

The Capability Constraint Binds

Certification requires $h_k^{raw} = 1{,}800$ clock hours of unassisted practice, so Equation (11) gives $h_k = 0.80 \times 1800 = 1{,}440$. With $\eta_k = 1.2$ and $B_k^{min} = 2{,}000$ hours, Equation (19) yields Equation (33):

$$B_k = \max\big(2000,\ 1.2 \times 24 \times 0.12 \times 1440\big) = 4{,}976.64 \text{ qualified-practice hours per year.} \tag{33}$$

The budget requires a mean credited share of $4976.64/16800 = 0.2962$, and Proposition 2 holds with about 18% to spare, at $\Phi_k \bar{\zeta}_k = 5{,}880$. The unconstrained allocation credits nothing and forms no successors.

Solving the Program and Reading Its Dual

Every inspection is treated as an identical copy of T_i. With one coupling constraint plus the assignment constraint, a vertex of the linear program (LP) relaxation has at most two active staffed modes, and the optimum (Table 5) sends 28.30% of inspections to $H{+}R/a_1$ and 71.70% to $H{+}A{+}R/a_1$: it staffs a developing inspector on *every* inspection. The dual follows from the tie between the two active modes, Equation (34):

$$0.308240 + 8\,\lambda_k(0.350) = 0.335452 + 8\,\lambda_k(0.275) \implies \lambda_k = \frac{0.027212}{0.6} = 0.045353, \tag{34}$$

so one additional qualified-practice hour costs 0.0454 objective units at the margin, the opportunity cost of tightening the constraint rather than the value of an hour of practice. Substituting λ_k into Equation (27)

Table 4. Worked bridge-inspection allocation. Normalized term valuations for the eight staffed modes, Civic Automation Debt from Equation (17), and Sustainable Civic Value from Equation (16). Roster a_0 is a certified lead alone; a_1 adds one developing inspector at an equal work split. Only D^{skill} differs between rosters.

	H		**H+A**		**H+R**		**H+A+R**	
Term	a_0	a_1	a_0	a_1	a_0	a_1	a_0	a_1
Flow terms, measured on this execution window								
Q service quality	0.72	0.69	0.86	0.83	0.80	0.77	0.91	0.88
S safety	0.55	0.53	0.60	0.58	0.88	0.86	0.90	0.88
P productivity	0.40	0.35	0.62	0.57	0.70	0.65	0.88	0.83
Eq^{srv} service equity	0.70	0.70	0.72	0.72	0.70	0.70	0.74	0.74
Tr trust	0.80	0.78	0.74	0.72	0.72	0.70	0.68	0.66
$Cost$ operating cost	0.85	0.95	0.62	0.72	0.58	0.68	0.50	0.60
En energy	0.20	0.20	0.28	0.28	0.55	0.55	0.60	0.60
Pr privacy risk	0.10	0.10	0.25	0.25	0.30	0.30	0.38	0.38
Developmental shares and stock-change terms								
ϕ_m human share	1.00	1.00	0.75	0.75	0.70	0.70	0.55	0.55
ψ_a developmental eligibility	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50
$\phi_m\psi_a$ credited share	0.000	0.500	0.000	0.375	0.000	0.350	0.000	0.275
$D^{skill} = 1 - \phi_m\psi_a$	1.00	0.50	1.00	0.625	1.00	0.650	1.00	0.725
D^{fall} fallback erosion	0.00	0.00	0.15	0.15	0.25	0.25	0.35	0.35
D^{acct} accountability dilution	0.00	0.00	0.07	0.07	0.05	0.05	0.13	0.13
D^{dep} vendor dependency	0.00	0.00	0.30	0.30	0.35	0.35	0.50	0.50
D^{trans} transition burden	0.00	0.00	0.10	0.10	0.20	0.20	0.30	0.30
$CAD_{i,m,a}$ from (17)	0.2800	0.1400	0.3816	0.2766	0.4260	0.3280	0.5004	0.4234
$SCV_{i,m,a}$ from (16)	0.2324	0.2391	0.2783	0.2773	0.3108	0.3082	**0.3426**	0.3355
Survives $S_i^{min} = 0.63$?	no	no	no	no	yes	yes	yes	yes

reproduces the optimum's ranking, Equation (35):

$$\begin{aligned} \widetilde{SCV}_{H+R/a_1} = \widetilde{SCV}_{H+A+R/a_1} &= 0.435229 \\ > \widetilde{SCV}_{H+A+R/a_0} &= 0.3426 \\ > \widetilde{SCV}_{H+R/a_0} &= 0.3108. \end{aligned} \tag{35}$$

The binding constraint reverses the winner first on *who is on the team* and only second on *which machines are used*. Integrally, 594 tasks in $H{+}R/a_1$ miss the budget by 0.24 hours and 595 meet it, so the integer optimum's mean SCV of 0.327742 lies 7.8×10^{-6} below the LP value of 0.327750.

Table 5. All two-mode mixes of admissible staffed modes that reach the required mean credited share of 0.2962. No pair excluding roster a_1 reaches a positive credited share. Of the single modes, $H{+}R/a_1$ alone meets the share at a mean SCV of 0.3082. Bold marks the optimum.

Mix	**Shares of tasks**	**Mean** $\phi_m\psi_a$	**Mean** SCV
$H{+}R/a_1$ with $H{+}A{+}R/a_1$	28.30% / 71.70%	0.2962	**0.3277**
$H{+}R/a_1$ with $H{+}A{+}R/a_0$	84.64% / 15.36%	0.2962	0.3135
$H{+}R/a_1$ with $H{+}R/a_0$	84.64% / 15.36%	0.2962	0.3086

What Preservation Costs

Unconstrained allocation yields a mean SCV of 0.3426 per task, or 719.5 objective units a year; the constrained optimum yields 0.3277, or 688.3 units. Preservation costs 31.2 units a year, 4.34% of annual objective value (0.33–9.60% across the sweep of Section 6.1), an average of 0.0063 units per protected

hour against the marginal 0.0454, as convexity requires. The city pays mainly in operating cost, whose normalized score rises 24.5% because every inspection carries a second person, and in productivity, down 11.5%, with smaller losses in quality and safety, which stays above its floor (Appendix S5, Table S7). Meeting the budget forms $4976.64/1440 = 3.456$ certified-equivalent practitioners a year against attrition of $24 \times 0.12 = 2.88$, a net gain of 0.576 set by the choice $\eta_k = 1.2$.

Retention does not bind in this domain. At the testbed's illustrative retention parameters ($\vartheta_0 = 0.30$, $\vartheta^{comp} = 0.70$, $\vartheta^{res} = 0.45$, $\tau^{ret} = 120$ working days), an inspector with twice the certification practice has $\vartheta^{\infty} = 0.8714$ and, by Proposition 1, a recency interval of 62.7 working days, whereas each of the 24 leads performs a human-led inspection every 2.86 working days on average under either allocation, because the policy twin keeps a human in every admissible mode. Retention binds where machine-only modes are admissible, which is what the reference execution examines.

Pipeline Arithmetic and the Required Trainee Cohort

The optimum staffs a developing inspector on all 2,100 inspections, 21,000 trainee clock hours a year, each of which accrues only $4976.64/21000 = 0.237$ qualified-practice hours, because practice is diluted by machine assistance ($\bar{\phi}_k = 0.5925$) and by sharing the work with a lead ($\bar{\omega}_k = 0.5$). At $\Theta_k = 1{,}500$ clock hours per trainee-year, Proposition 3 gives Equations (36) and (37):

$$\tau_k \;=\; \frac{1800}{1500} \cdot \frac{1}{0.5925 \times 0.5} \;=\; 1.20 \times 3.376 \;=\; 4.05 \text{ years}, \tag{36}$$

$$\hat{n}_k \;=\; 1.2 \times 24 \times 0.12 \times 4.05 \;=\; 14.0 \text{ developing practitioners}. \tag{37}$$

Two routes agree: 21,000 trainee clock hours at 1,500 per trainee-year is 14.0 posts, and Little's law at 3.456 completions a year and 4.05 years' residence gives the same. Certification hours imply 1.2 years; under the optimal mix the same practitioner needs 4.05, 3.38 times longer (1.69 from machine assistance, 2.0 from sharing the work), so a cohort sized from certification hours provisions four trainees where fourteen are required (Figure 3). The dilution lives in $\bar{\phi}_k$ and $\bar{\omega}_k$, properties of the *allocation* rather than the workforce, which is why a headcount plan does not show it. If a scheme counts hours worked beside a lead, $\bar{\omega}_k$ leaves the factor, τ_k falls to 2.03 years, and $\hat{n}_k$ to 7.0.

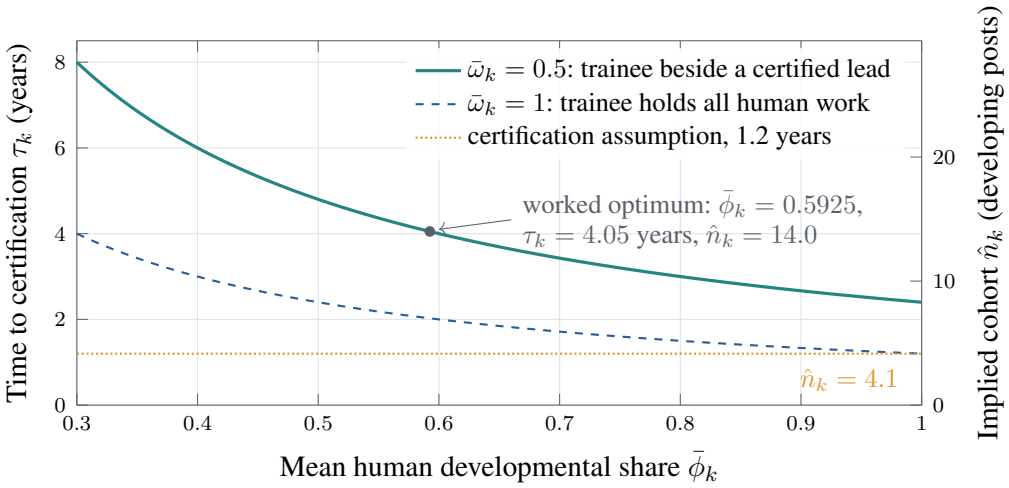


Figure 3. Time to certification under Equation (30) against the mean human developmental share of the selected staffed modes, at the worked domain's solo-practice time $h_k^{raw}/\Theta_k = 1800/1500 = 1.2$ years. The solid curve is a developing practitioner holding half of a roster's human work, as at the worked optimum; the dashed curve is one holding all of it. The certification figure of 1.2 years is reached only for unshared practice without machine assistance. The right axis is the steady-state cohort of Equation (31), $\hat{n}_k = 3.456\,\tau_k$. Every point is Equation (30) evaluated at the stated values.

Sensitivity and Feasibility

Table 6 sweeps the six load-bearing parameters one at a time. Staffing developing practitioners holds at every feasible value, but its magnitude does not: in the eight feasible rows labeled "no reversal" the optimum keeps the lead-only roster on 1.3–46.1% of tasks and the dual falls to 0.0005–0.0033. The mode reversal occurs exactly where the required credited share B_k/Φ_k exceeds what $H{+}A{+}R/a_1$ delivers (0.275), so it depends on both sides of that inequality. Feasibility is fragile: the baseline sits at 85% of the critical value on all three axes Proposition 2 bounds, so a stricter licensing regime ($h_k^{raw} = 2{,}400$) or a retirement wave ($r_k = 0.16$) makes the budget unattainable. The cost of preservation ranges from 0.33% to 9.60%. Two parameter shocks, a feasible retirement wave at $r_k = 0.14$ and an infeasible one at $r_k = 0.25$ whose deficit of 4,488 hours no allocation can close, are worked in Appendix S5.2.

Table 6. One-at-a-time sensitivity of the worked allocation, by exact arithmetic. Baseline rows in bold. "Infeasible" means Proposition 2 fails. "No reversal" means the optimum keeps $H{+}A{+}R$ on every task and mixes its two rosters; "reversal" means $H{+}R$ enters the optimum.

Parameter	Value	Optimum	λ_k	Cost	Status
h_k^{raw} (clock h)	1200	28.2% $H{+}A{+}R/a_0$ + 71.8% $H{+}A{+}R/a_1$	0.0033	1.50%	no reversal
	1500	10.2% $H{+}A{+}R/a_0$ + 89.8% $H{+}A{+}R/a_1$	0.0033	1.88%	no reversal
	1800	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	2127	boundary of Proposition 2	—	—	critical
	2400	—	—	—	infeasible
	3000	—	—	—	infeasible
r_k (per year)	0.06	46.1% $H{+}A{+}R/a_0$ + 53.9% $H{+}A{+}R/a_1$	0.0033	1.13%	no reversal
	0.09	19.2% $H{+}A{+}R/a_0$ + 80.8% $H{+}A{+}R/a_1$	0.0033	1.69%	no reversal
	0.12	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.14	94.1% $H{+}R/a_1$ + 5.9% $H{+}A{+}R/a_1$	0.0454	9.57%	near-critical
	0.1418	boundary of Proposition 2	—	—	critical
η_k	0.8	28.2% $H{+}A{+}R/a_0$ + 71.8% $H{+}A{+}R/a_1$	0.0033	1.50%	no reversal
	1.2	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	1.4	94.1% $H{+}R/a_1$ + 5.9% $H{+}A{+}R/a_1$	0.0454	9.57%	near-critical
	1.6	—	—	—	infeasible
ϕ_{H+R}	0.55	—	—	—	infeasible
	0.60	84.9% $H{+}R/a_1$ + 15.1% $H{+}A{+}R/a_1$	0.1515	9.60%	reversal
	0.70	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.85	14.2% $H{+}R/a_1$ + 85.8% $H{+}A{+}R/a_1$	0.0188	3.02%	reversal
ϕ_{H+A+R}	0.45	57.0% $H{+}R/a_1$ + 43.0% $H{+}A{+}R/a_1$	0.0241	7.00%	reversal
	0.55	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.60	1.3% $H{+}A{+}R/a_0$ + 98.7% $H{+}A{+}R/a_1$	0.0023	1.62%	no reversal
	0.70	15.4% $H{+}A{+}R/a_0$ + 84.6% $H{+}A{+}R/a_1$	0.0009	0.63%	no reversal
ψ_{a_1} (supervision)	0.40	—	—	—	infeasible
	0.4232	boundary of Proposition 2	—	—	critical
	0.50	**28.3%** $H{+}R/a_1$ **+ 71.7%** $H{+}A{+}R/a_1$	**0.0454**	**4.34%**	**baseline**
	0.67	19.6% $H{+}A{+}R/a_0$ + 80.4% $H{+}A{+}R/a_1$	0.0005	0.33%	no reversal

6.2 Reference Execution of the Protocol

Table 7 reports the five metric families for every policy and scenario, Table 8 the eight pre-specified hypotheses, and Figure 4 the number of current practitioners over the horizon. Appendix S1.5 gives every metric with its 95% interval. Four of the eight hypotheses are supported, comprising 7 of the 12 tests; the other four are not, and two of those fail in the opposite direction to the prediction.

Skill Retention and Formation

Retention is where the policies differ most. After the first week every policy has between 31.3 and 31.4 current practitioners on average, of 32 initially certified. Under efficiency-focused allocation, the sanitation and casework practitioners whose work moves to robots and AI agents lapse within about a year, and the count falls to 9.6 by the end of the horizon (Figure 4); static allocation, whose fixed table also sends that

Table 7. Reference execution: means over 30 paired seeds per policy and scenario. "No learning" and "No retention" are the CivicWorkOS ablations. The cost index divides each policy's cost per completed task by the efficiency-focused value on the same seed. A practitioner is current when true proficiency is at least ϑ^{comp}. Window rows count tasks arriving in, or faults occurring in, the event window. Appendix S1.5 gives 95% intervals for every entry.

Metric	Static	Efficiency	CivicWorkOS	No learning	No retention
Routine operations (three years)					
Task completion (%)	99.4	99.3	97.8	99.5	96.2
Service cost index (efficiency = 100)	123.6	100.0	173.4	180.2	159.0
Overloaded practitioner-days (%)	0.11	4.18	0.04	6.49	0.04
Current share of certified, end	0.56	0.42	0.84	0.84	0.43
Current practitioners, end	14.2	9.6	27.5	28.4	14.1
Certified practitioners, end	25.7	23.0	32.8	33.9	32.9
Certifications over the horizon	2.8	0.0	10.9	12.3	10.9
Takeover success, all faults	0.876	0.918	0.893	0.897	0.790
Incidents per 1,000 tasks	7.83	6.32	5.57	6.22	5.45
Proficiency estimate error (MAE), end	0.023	0.020	0.023	0.035	0.028
Demand surge (30-day window)					
Completion of window tasks (%)	88.2	87.7	92.7	96.6	85.6
Overloaded practitioner-days (%)	0.11	3.09	0.04	5.60	0.03
Automation disruption (20-day window)					
Completion of window tasks (%)	76.4	26.8	71.9	93.5	52.4
Takeover requests in window	306.3	23.5	36.7	290.5	32.0
Takeover success in window	0.874	0.734	0.812	0.878	0.720
Incidents per 1,000 window tasks	35.8	3.8	5.0	31.0	3.6
Reliability estimate error in window (MAE)	0.403	0.131	0.145	0.403	0.147

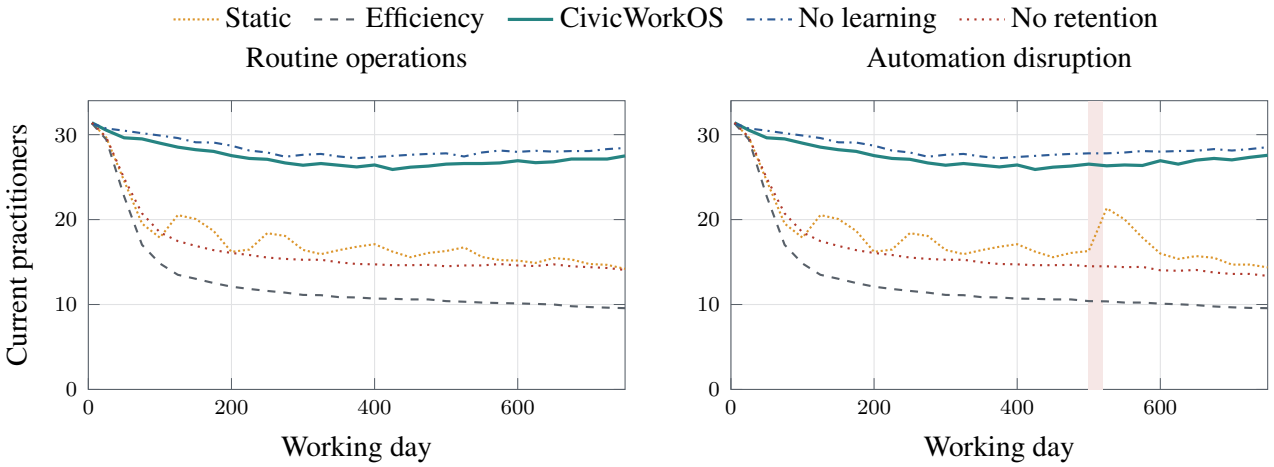


Figure 4. Number of certified practitioners who are current (true proficiency at least ϑ^{comp}), averaged over 30 seeds, in the routine (left) and disruption (right) scenarios; the shaded band is the disruption window. All policies start with the same 32 certified practitioners across the three domains. Under efficiency-focused and static allocation, practitioners whose work has moved to machines lapse within the first year; CivicWorkOS keeps them current and adds newly certified practitioners. Removing periodic human-led tasks (No retention) loses most of that advantage even though formation is unchanged.

work to machines, ends at 14.2. CivicWorkOS ends with 27.5 current practitioners and a current share of 0.84, against 0.42 and 0.56, so H1 is supported against both comparators with differences of 0.42 [0.38, 0.46] and 0.28 [0.24, 0.32]. Formation is the second half: CivicWorkOS certifies 10.9 practitioners over three years where efficiency-focused allocation certifies none, and ends with 9.8 [9.0, 10.7] more certified practitioners (H2).

Table 8. Pre-specified hypotheses and outcomes. Each test compares CivicWorkOS with the named policy on 30 paired seeds. "Estimate" is the mean paired difference (CivicWorkOS minus comparator), or for H4 the mean cost ratio, with its 95% interval. p_{Holm} is the Holm-adjusted two-sided Wilcoxon signed-rank p-value over the 12 tests; r is the matched-pairs rank-biserial correlation. The decision rule is stated in Section 5.4.

ID	Hypothesis	Scenario	vs.	Criterion	Estimate [95% CI]	p_{Holm}	r	Verdict
H1	Retention: current share at end is higher	routine	Efficiency	≥ 0.15	+0.420 [+0.380, +0.460]	$< 10^{-7}$	+1.00	supported
H1	Retention: current share at end is higher	routine	Static	≥ 0.15	+0.280 [+0.238, +0.322]	$< 10^{-7}$	+1.00	supported
H2	Formation: more certified practitioners at end	routine	Efficiency	sign	+9.83 [+8.99, +10.68]	$< 10^{-7}$	+1.00	supported
H3	Takeover success in the disruption window is higher	disruption	Efficiency	≥ 0.10	+0.078 [+0.036, +0.120]	2.3e−3	+0.66	not supported
H3	Takeover success in the disruption window is higher	disruption	Static	≥ 0.10	−0.062 [−0.100, −0.024]	2.3e−3	−0.63	not supported
H4	Cost per completed task at most 115% of efficiency	routine	Efficiency	ratio ≤ 1.15	1.734 [1.664, 1.804]	$< 10^{-7}$	+1.00	not supported
H5	Completion within 2 points of efficiency	routine	Efficiency	margin −0.02	−0.016 [−0.026, −0.005]	1.2e−4	−0.80	not supported
H6	Overload share is lower	routine	Efficiency	sign	−0.041 [−0.050, −0.033]	$< 10^{-7}$	−1.00	supported
H6	Overload share is lower	surge	Efficiency	sign	−0.031 [−0.038, −0.023]	$< 10^{-7}$	−1.00	supported
H6	Overload share is lower	disruption	Efficiency	sign	−0.043 [−0.052, −0.035]	$< 10^{-7}$	−1.00	supported
H7	Learning raises disruption-window completion	disruption	No learning	sign	−0.215 [−0.246, −0.185]	$< 10^{-7}$	−1.00	not supported
H8	Retention mechanism raises current share	routine	No retention	≥ 0.10	+0.411 [+0.378, +0.444]	$< 10^{-7}$	+1.00	supported

The retention ablation separates the two. Without periodic human-led tasks and refreshers, CivicWorkOS forms as many practitioners (10.9 certifications, 32.9 certified at the end) but keeps only 14.1 of them current, a current share of 0.43, so H8 is supported with a difference of 0.41 [0.38, 0.44]. Formation without

retention fills the register with certified practitioners who are not ready, and the gap between *certified* and *current* that municipal records cannot see is, in this testbed, larger than any difference in formation.

Workload

CivicWorkOS uses its practitioners far more than efficiency-focused allocation does (mean workload index 0.41 against 0.15), yet it overloads them least: 0.04% of practitioner-days exceed the ceiling, against 4.18% under efficiency-focused allocation, which plans human work up to the hours cap without regard to individual workload responses. H6 is supported in all three scenarios. The no-learning ablation is the informative contrast: it enforces the same ceiling but plans with every practitioner's workload response fixed at the nominal value, and it overloads on 6.49% of practitioner-days, the most of any policy. The ceiling protects practitioners only when it is evaluated on learned, individual workload responses, which in this testbed come from the physiological channel.

Cost and Completion

Preservation is expensive, and the protocol says so. CivicWorkOS's cost per completed task is 173.4% [166.4, 180.4] of efficiency-focused allocation, far above the 115% margin of H4, and its routine completion is 97.8% against 99.3%, a difference of -1.6 points [-2.6, -0.5] whose interval crosses the 2-point margin, so neither non-inferiority hypothesis is supported. The cost comes from wages for supervised and human-led work in domains where a machine hour costs 0.08 to 0.30 units against 1.0 for a practitioner hour (Table 3), so the ratio scales with that price gap. It is a different quantity from the 4.34% of Section 6.1, which is the loss in a normalized multi-term objective. The completion loss arises when the workload ceiling and the oversight gate hold tasks that no admissible human roster can absorb before the deadline.

Accuracy of the Learned Estimates

Learning makes the proficiency estimates more accurate: at the end of the routine horizon their mean absolute error (MAE) is 0.023 under CivicWorkOS against 0.035 for the no-learning ablation, which relies on the nominal model and the practice log alone. For machine reliability the picture depends on the scenario. In routine operations, where true reliability barely leaves its commissioning value, the learned estimate is noisier (error 0.030) than simply trusting that value (0.010), the price of a belief that can move. In the disruption window the learned estimate tracks the failure, with an error of 0.145 against 0.403 for the policies that cannot update.

Demand Surge

The retained workforce is also surge capacity. In the 30-day window of doubled demand, CivicWorkOS completes 92.7% of arriving tasks against 87.7% for efficiency-focused and 88.2% for static allocation, while keeping overload at 0.04% against 3.09% under efficiency-focused allocation. The no-learning ablation completes more (96.6%) by overloading practitioners on 5.60% of days.

Automation Disruption and Control Transfer

The disruption scenario exposes a trade-off that the pre-specified completion metric was not designed to capture. The two policies that learn detect the silent degradation through failures and telemetry and stop dispatching work to the failing machines: in the 20-day window they raise 36.7 (CivicWorkOS) and 23.5 (efficiency-focused) takeover requests, against 306.3 and 290.5 for static allocation and the no-learning ablation, which keep dispatching. The consequence for safety is large: 5.0 and 3.8 incidents per 1,000 window tasks against 35.8 and 31.0. The consequence for completion runs the other way. Efficiency-focused allocation completes only 26.8% of window tasks, because the practitioners to whom work must shift have lapsed and fail the safety floor. CivicWorkOS, whose practitioners are current, completes 71.9%, 2.7 times as many, but fewer than the no-learning ablation (93.5%), which keeps pushing work through failing machines and recovers it with successful takeovers by current practitioners, at six times CivicWorkOS's incident rate. H7, that learning raises window completion, therefore fails in the opposite direction (-0.215 [-0.246, -0.185]).

Takeover success shows the same structure. CivicWorkOS recovers 0.812 of its window faults against 0.734 for efficiency-focused allocation, a difference of 0.078 [0.036, 0.120] that is statistically significant after Holm adjustment but below the 0.10 threshold of H3, and below static allocation (0.874), so H3 is not supported. When the readiness set is empty, CivicWorkOS commands a minimal-risk stop rather than

hand control to an unready or overloaded practitioner, and the metric counts that stop as unrecovered; it commands 29.4 such stops per run in the disruption scenario, where static allocation, which hands every takeover to someone, commands none. Removing retention lowers window takeover success to 0.720 and window completion to 52.4%: readiness for takeover is produced by the periodic human-led tasks of the preceding months.

Exploratory Analyses

Two analyses were added after the reference execution to answer questions it left open. They are exploratory: they are not part of the pre-specified family, they reuse the 30 reporting seeds so that comparisons stay paired, and their intervals are not adjusted for multiplicity (Table 9).

Table 9. Exploratory analyses, added after the reference execution and not pre-specified; 30 paired seeds, unadjusted 95% intervals. Panel A compares CivicWorkOS with a variant that learns from task outcomes only, every multimodal channel (skill probes, telemetry, physiological workload readings) removed. Panel B repeats the comparison of CivicWorkOS with efficiency-focused allocation under a halved and a doubled retention time constant and under tripled machine prices. MAE is the mean absolute error of the allocator's estimate against the true latent value.

A. Multimodal channels removed (learning from task outcomes only)

Metric	**CivicWorkOS**	**Outcomes only**	**Difference [95% CI]**
Task completion (%)	97.8	87.0	+10.8 [+8.7, +12.9]
Overloaded practitioner-days (%)	0.04	7.74	−7.70 [−9.03, −6.37]
Current practitioners, end	27.5	21.3	+6.2 [+4.4, +8.1]
Proficiency estimate error (MAE), end	0.023	0.096	−0.073 [−0.078, −0.068]
Completion of window tasks (%) (disruption)	71.9	56.5	+15.4 [+11.5, +19.3]
Takeover success in window (disruption)	0.812	0.366	+0.446 [+0.363, +0.528]
Incidents per 1,000 window tasks (disruption)	5.0	2.3	+2.7 [+2.0, +3.3]
Reliability estimate error in window (MAE) (disruption)	0.145	0.269	−0.124 [−0.136, −0.112]

B. Sensitivity, routine operations: CivicWorkOS minus efficiency-focused

Variant	**Current share, end**	**Completion (points)**	**Cost index (efficiency = 100)**
Baseline	+0.42 [+0.38, +0.46]	−1.6 [−2.6, −0.5]	173.4 [166.4, 180.4]
$\tau^{ret} \times 0.5$	+0.46 [+0.42, +0.50]	−1.4 [−2.5, −0.3]	176.9 [169.4, 184.5]
$\tau^{ret} \times 2$	+0.38 [+0.34, +0.41]	−1.0 [−1.4, −0.6]	175.2 [171.7, 178.8]
Machine prices ×3	+0.35 [+0.30, +0.39]	−1.8 [−2.7, −0.9]	144.3 [139.1, 149.4]

What the multimodal channels contribute. Learning from outcomes alone degrades nearly every metric. Routine completion falls from 97.8% to 87.0%, overload rises from 0.04% to 7.74% of practitioner-days, 6.2 fewer practitioners are current at the end of the horizon, and in the disruption window takeover success falls from 0.812 to 0.366. The cause is identifiability. One outcome carries information about the lead, the trainee, the machine, and the requirement at once; without a direct observation of any of them the beliefs drift, practitioners are wrongly estimated to have lapsed, the readiness set empties, and work that only a current human may perform waits. The error of the proficiency estimates at the end of the horizon rises from 0.023 to 0.096, worse than the 0.035 of the no-learning ablation, which never updates at all: confounded outcome updates do not merely fail to help, they mislead. The variant also has fewer window incidents (2.3 against 5.0 per 1,000 tasks), because it attempts less of the work. The channels are idealized (Section 3.4), so the size of their contribution here is an upper bound on what real sensors would deliver; its direction shows that the learning mechanism depends on observations beyond task outcomes, which is the case for multimodal sensing in this setting.

Robustness to the retention constant and to prices. Across the baseline and the three variants (halving or doubling τ^{ret} and tripling machine prices), the retention advantage of CivicWorkOS over efficiency-focused allocation stays between 0.35 and 0.46 in current share, and its completion penalty between 1.0 and 1.8 points. The cost index moves with prices, from 173.4 at baseline to 144.3 when machine hours cost three times as much: the cost of preservation is set by the price of human relative to machine hours, while its benefit for retention is not.

7 DISCUSSION

7.1 What the Results Establish, and What They Do Not

The worked allocation establishes by arithmetic that the capability constraints are computable, that they bind, that their dual is legible, and how the trainee cohort a domain needs depends on the allocation itself. The reference execution establishes that, in a synthetic testbed with stated parameters, the mechanisms behave as specified: preservation sustains current practitioners, forms new ones, keeps workload under the ceiling, and provides surge capacity, at a substantial cost in money and a small one in routine completion. Neither shows that any input is correct for a real city. The testbed's parameters are illustrative, its proficiency model is assumed rather than fitted, and its practitioners are simulated; the elicitation protocol of Appendix S6 has not been run. The results are evidence about the mechanisms, not about municipalities, and the pre-specified hypotheses that failed (H3, H4, H5, H7) are as much a part of that evidence as those that held.

Part of the retention result is true by construction, and we separate that part from the rest. The retention constraint schedules human-led tasks so that the decay model does not cross the competence level, and H1 and H8 then measure whether it crossed; under the assumed dynamics, a policy that schedules practice this way keeps practitioners current almost by definition. What the execution adds is that the schedule is *attainable* inside the workload ceiling and the safety floor, with learned rather than true proficiency, while other work proceeds, and what it costs to attain. The other findings are not built in: whether formation and retention together supply surge capacity, how learning changes completion and safety under disruption, whether the readiness rule improves takeover, and the size of the cost and completion penalties all emerge from the interaction of the mechanisms with the task streams.

Against the research questions: RQ1 and RQ2 are answered by specification (Sections 3 and 4) and by the behavior of the learning layer in the execution; RQ3 analytically for one domain (Propositions 1–3, Section 6.1) and by the retention and formation results; RQ4 by the readiness rule and the disruption results, which show both its benefit over efficiency-focused allocation and its cost in unrecovered takeovers; and RQ5 by the protocol itself, which another group can run unchanged.

7.2 Certification Is Not Readiness

The clearest finding concerns the difference between being certified and being current. Records-based competence, the hours a practitioner has logged, does not decay; proficiency does. An allocation policy that moves routine work to machines leaves the register of certified staff intact while emptying it of people ready to supervise or take over, and nothing in the register shows this. The no-retention ablation makes the point sharply: it forms practitioners as fast as full CivicWorkOS and still ends with fewer current practitioners than static allocation. Proposition 1 gives the remedy its shape. Keeping a practitioner current requires a human-led task at least every $\bar{e}_w$ days, which costs few hours but cannot be banked, so it must be scheduled, as aviation's recency rules schedule it (Federal Aviation Administration, 2024). It also implies that newly certified practitioners, whose attainable proficiency sits at the competence level, have no recency margin at all until further practice consolidates it.

7.3 Learning, Safety, and What the Protocol Measures

The failure of H7 is the most instructive result. Adaptive learning did what it should: it detected the degraded machines and stopped using them, cutting window incidents roughly sixfold relative to the no-learning ablation. But the work those machines had carried had to go somewhere, and the human capacity available under a workload ceiling was smaller than the load. A policy that cannot see the degradation completes more tasks, and causes more incidents, because it keeps using machines whose machine-only modes still succeed about a third of the time, and it has current practitioners to catch their failures. Which outcome a city prefers is a value judgment about service continuity against safety, and a protocol that scores completion alone will reward not learning. We pre-specified completion and report it; we recommend that future protocols pre-specify a joint criterion, such as completion subject to an incident ceiling. For deployment the lesson is that learning shifts load onto people at exactly the moment automation fails, so the human reserve of Equation (22) must be sized for that shift, not only for the steady state.

Development surfaced a second hazard specific to learning human state. Before the refresher route was added, practitioners whose proficiency estimate dipped below the competence level were excluded

from human-led work, lost practice, and lapsed in fact, so the estimate produced the decline it predicted (Appendix S1.4). Any allocator that learns human proficiency and acts on it closes this loop. The supervised refresher breaks it by giving a practitioner estimated to have lapsed a safe route back to practice, and we expect some such route to be necessary in any learning allocator that conditions assignment on estimated skill.

7.4 Implications for Data-Driven Neurorobotics and Human–Robot Interaction

For human–robot interaction, the work suggests treating the human side of the team as a learned, non-stationary state with its own dynamics, not as a fixed capability profile. The belief model is the minimal version of that idea: interpretable and updated in closed form. The results show that its value depends on what it observes. With multimodal channels, its proficiency estimates carry about a third less error than the nominal model's and the workload ceiling protects practitioners; with task outcomes alone, its estimates become worse than not learning and every downstream metric suffers (Table 9). For data-driven human–robot interaction, that is an argument for designing the sensing alongside the allocator, and for treating identifiability, not only accuracy, as a requirement of any learner that conditions work on estimated human state. The protocol is the other contribution the field can use directly. Its scenarios, metrics, and decision rules apply to any allocation policy, learned or hand-written, that can be expressed as a selection rule and a takeover rule in the released testbed, and its seeds and common random numbers make comparisons paired and reproducible. Natural extensions include richer learners for the same beliefs, such as hierarchical models that pool information across practitioners; federated estimation that keeps physiological data on the practitioner's device (Appendix S4.3); and calibration of the proficiency model against real practice logs and skill assessments.

7.5 Who Gains and Who Loses

Practitioners on the municipal payroll gain protected practice and, through the access constraint of Appendix S4, a published rule for who receives it. Residents gain resilience: a retained workforce absorbs surges and catches failing machines. The city pays, in the testbed about three-quarters more per completed task, and the panel that sets η_k, the recency guard, and the workload ceiling decides how much. Workers off the payroll, contracted, agency, or platform-mediated (De Stefano, 2016; Wood et al., 2019), are invisible to the ledger and protected by none of the constraints, so where they carry much of a city's sanitation and inspection labor, CivicWorkOS would preserve the permanent workforce while the contracted one absorbed the displacement. The framework also legitimates automation as much as it constrains it: a satisfied budget and a recorded decision can justify displacing workers once the budget is met. It is not neutral between these outcomes; it makes them explicit and contestable (Appendix S4.4).

7.6 Limitations

- *Synthetic evaluation.* The testbed's parameters are illustrative and its practitioners simulated; no result transfers to a city without calibration, and the cost and completion trade-offs depend on the price and capacity ratios of Table 3.
- *Assumed proficiency dynamics.* Exponential decay toward a residual level with full restoration by one human-led task is a simple member of the learn–forget family (Jaber and Bonney, 1996); partial restoration, interference, and skill-specific decay rates (Arthur et al., 1998) would change the recency interval.
- *Simple learner.* Independent Gaussian beliefs ignore correlations between practitioners and between skills, and the multimodal channels are modeled as unbiased noisy observations, which real physiological workload estimates are not (Dehais et al., 2020).
- *Metric choice.* Completion was pre-specified without a safety qualifier, which is why learning appears harmful under disruption (Section 7.3).
- *No regret guarantee.* Beyond Proposition 5 the online rule has no regret or cumulative-violation bound (Appendix S2.3), and the city-wide program's tractability is unmeasured.
- *Data protection.* Per-practitioner proficiency and physiological workload estimates are sensitive; the measures of Appendix S4.3 bound but do not remove the exposure, and participation in monitoring must be voluntary.

- *Formal-employment presumption.* The framework preserves the visible payroll workforce and cannot see the rest. This is the limitation we would fix first.

8 CONCLUSIONS

An allocation decides more than whether a human, an AI agent, or a robot performs a task better today. It decides who will be able to supervise, decide, and take over tomorrow. CivicWorkOS makes that second decision explicit: it treats human proficiency as a resource that practice builds and gaps erode, learns each participant's capability from outcomes and multimodal observations, protects supervised practice and periodic human-led work under workload limits, and transfers control of a failing machine only to a practitioner who is ready.

The worked bridge-inspection allocation shows by exact arithmetic that preserving capability reverses the efficient choice at a marginal cost of 0.0454 objective units per protected hour and 4.34% of objective value (0.33–9.60% across the sweep), and that machine assistance and shared supervision stretch formation from 1.2 to 4.05 years, so a cohort sized from certification hours provisions four trainees where fourteen are required. The reference execution of the protocol, on 30 paired seeds, shows that capability-preserving allocation ends a three-year horizon with 27.5 current practitioners against 9.6 under efficiency-focused allocation, certifies 10.9 new practitioners against none, and keeps overload at 0.04% of practitioner-days against 4.18%, at 1.73 times the cost per completed task and 1.6 points lower routine completion. It also shows that learning, by detecting failing automation, cuts incidents during a disruption roughly sixfold but completes fewer tasks than continuing to use the failing machines.

Future work should calibrate the proficiency model and the multimodal channels against real practice logs, skill assessments, and physiological recordings; pre-specify joint completion-and-safety criteria; prove regret and violation bounds for the online rule; extend the ledger to contracted and platform-mediated workers; and pilot the evidentiary record and readiness test against an existing municipal allocation practice, which a city could adopt before any optimizer.

ABBREVIATIONS

AI, artificial intelligence; CAD, Civic Automation Debt; EEG, electroencephalography; HRV, heart-rate variability; LP, linear program; MAE, mean absolute error; SCV, Sustainable Civic Value.

ETHICS STATEMENT

This study involved no human participants, no animal subjects, and no personal data. The worked allocation is analytical, and the reference execution runs on a synthetic testbed whose practitioners are simulated. The expert-elicitation protocol specified in Appendix S6 has not been executed; its execution would require institutional ethics approval and written informed consent, as Appendix S6.1 states. Ethical review and approval were therefore not required for this work in accordance with institutional and national requirements.

DATA AVAILABILITY STATEMENT

No empirical datasets were collected or analyzed in this study. The reference implementation, the synthetic testbed, the pre-specified hypotheses, the analysis scripts, and the summarized outputs of the reference execution and of the exploratory analyses are openly available at `https://github.com/aliakarma/CivicWorkOS`. Running `python scripts/run_protocol.py --paper` regenerates every simulation result reported here, including the per-run records, `python scripts/run_exploratory.py` regenerates the exploratory analyses, and `python Paper/Frontiers/audit_numbers.py` recomputes every number the article prints. The testbed's data are synthetic and must not be cited as measurements of any municipality.

AUTHOR CONTRIBUTIONS

TAS and AAk conceptualized the study and developed the research methodology. RMA supervised the study and provided guidance on the research design and interpretation of the findings. MTN and AAl contributed to the investigation, formal analysis, and validation. TAS and AAk developed the reference implementation, the evaluation testbed, and their verification, and prepared the visualizations. TAS prepared the initial manuscript. All authors contributed to the revision and editing of the manuscript, critically reviewed the final version, and approved the submitted version. AAk denotes Ali Akarma and AAl Abdulaziz Alqurashi.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

GENERATIVE AI STATEMENT

The authors declare that generative AI was used in the preparation of this manuscript. An AI assistant (Claude, Anthropic) was used to help draft and edit text and to implement and test parts of the reference code. The authors reviewed and verified all content, code, and results, and take full responsibility for the manuscript.

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